Restore initial image system

Restore initial image system

- 1. Format the SSD
  - 1.1. Download DiskGenius
  - 1.1. Use DiskGenius
    - 1.1.1, Delete partition
    - 1.1.2. Create a new partition
- 2. Restore the factory image
  - 2.1. Install Win32DiskImager
  - 2.2. Use Win32DiskImager
- 3. Description

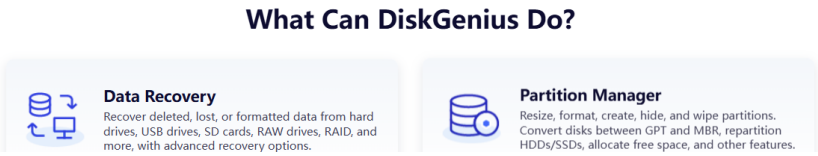
Note: Users must prepare an SSD enclosure in advance. Install the SSD into the enclosure, connect it to a computer, and proceed with formatting and system flashing. If you are unsure how to remove the SSD, please refer to the steps outlined in the "02.Jetson basic course: 05. Replacing the SSD" lesson to safely detach the drive before installing it into the enclosure.

1. Format the SSD

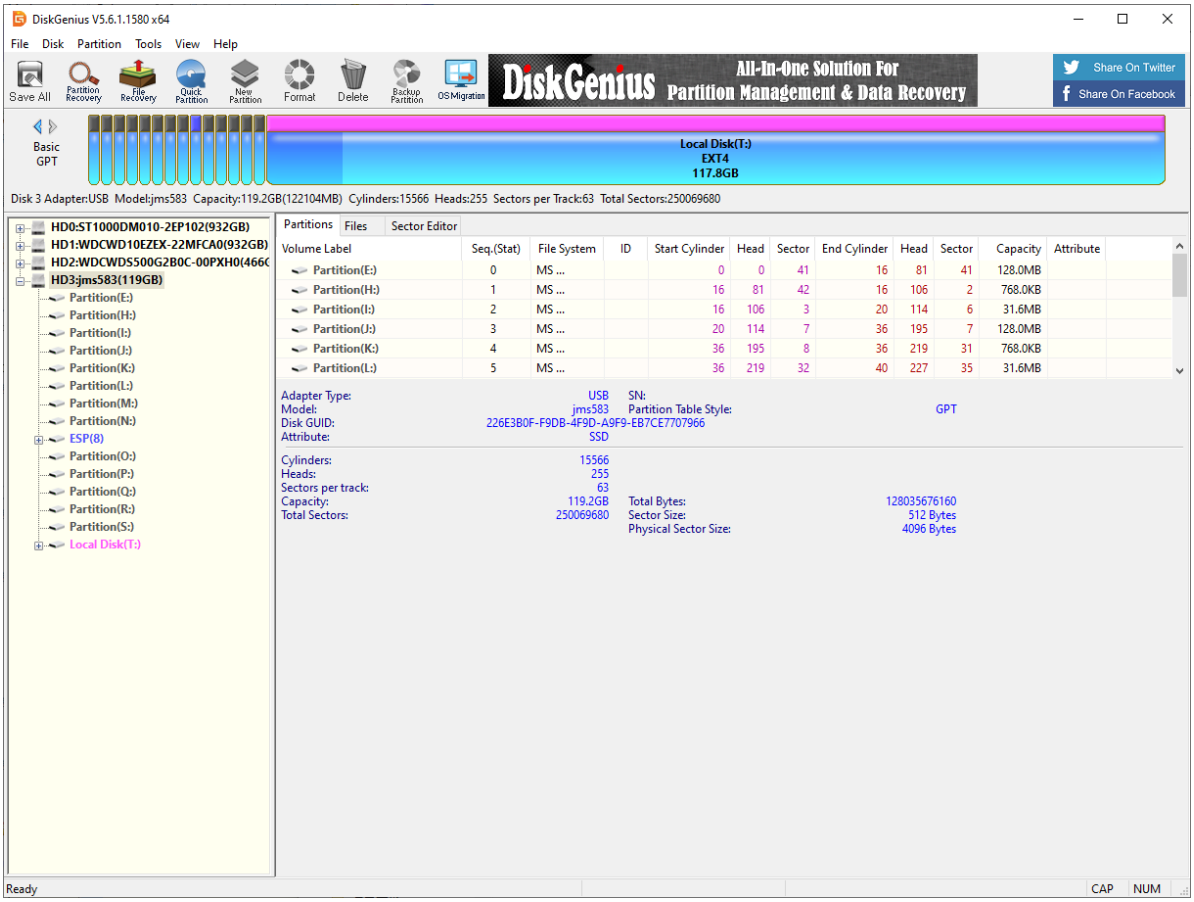
Before restoring the factory image, you need to format the SSD into exFAT format.

1.1. Download DiskGenius

Download URL: https://www.diskgenius.com/



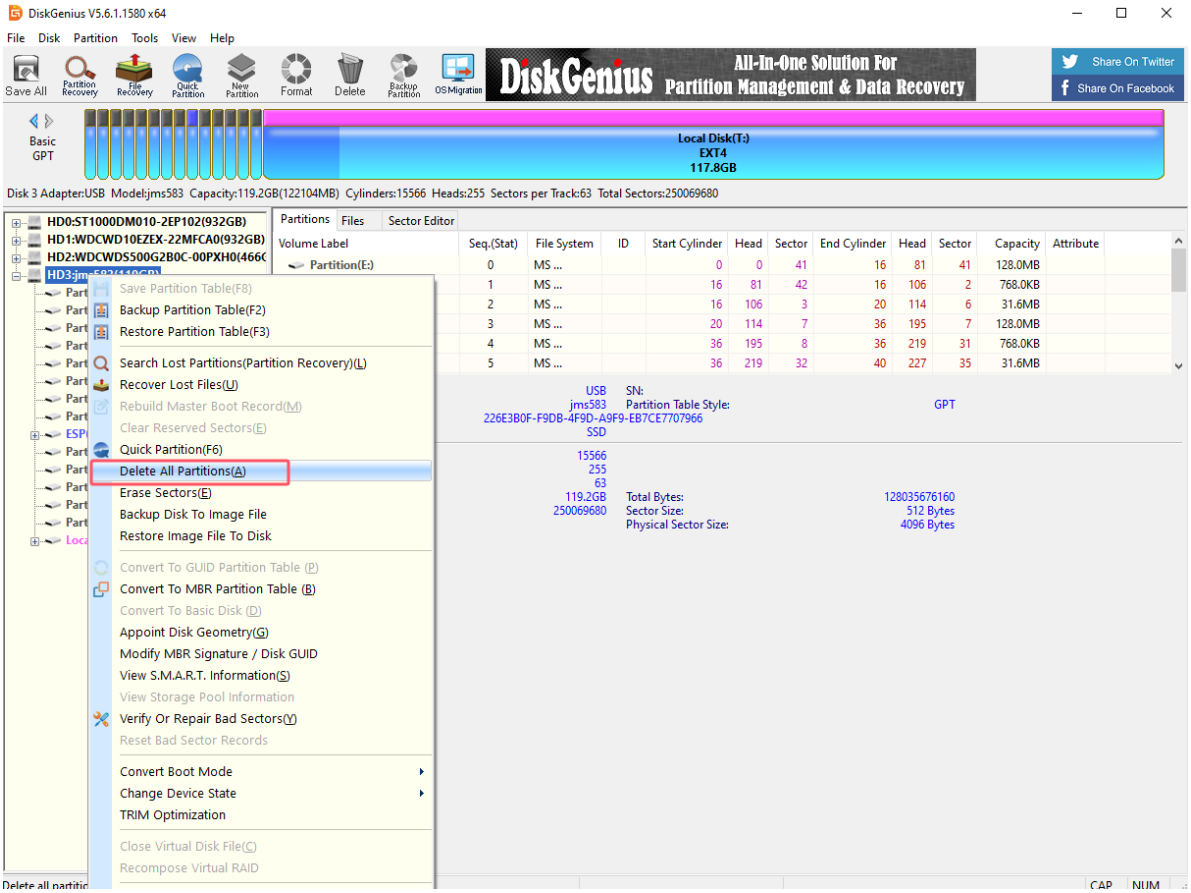
Double-click the exe file you just downloaded to install DiskGenius. Follow the prompts to install the software on the Windows computer. After opening the software, it will be as shown below.

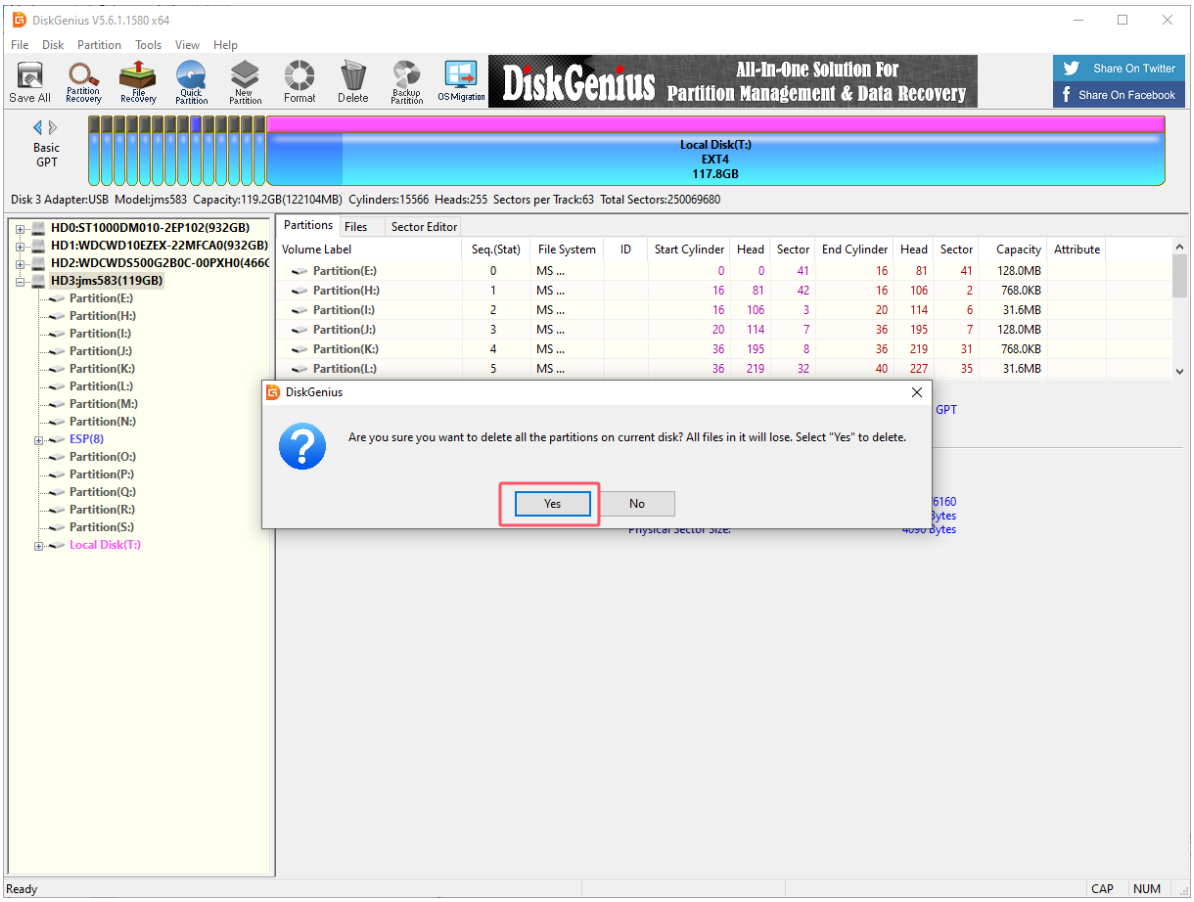


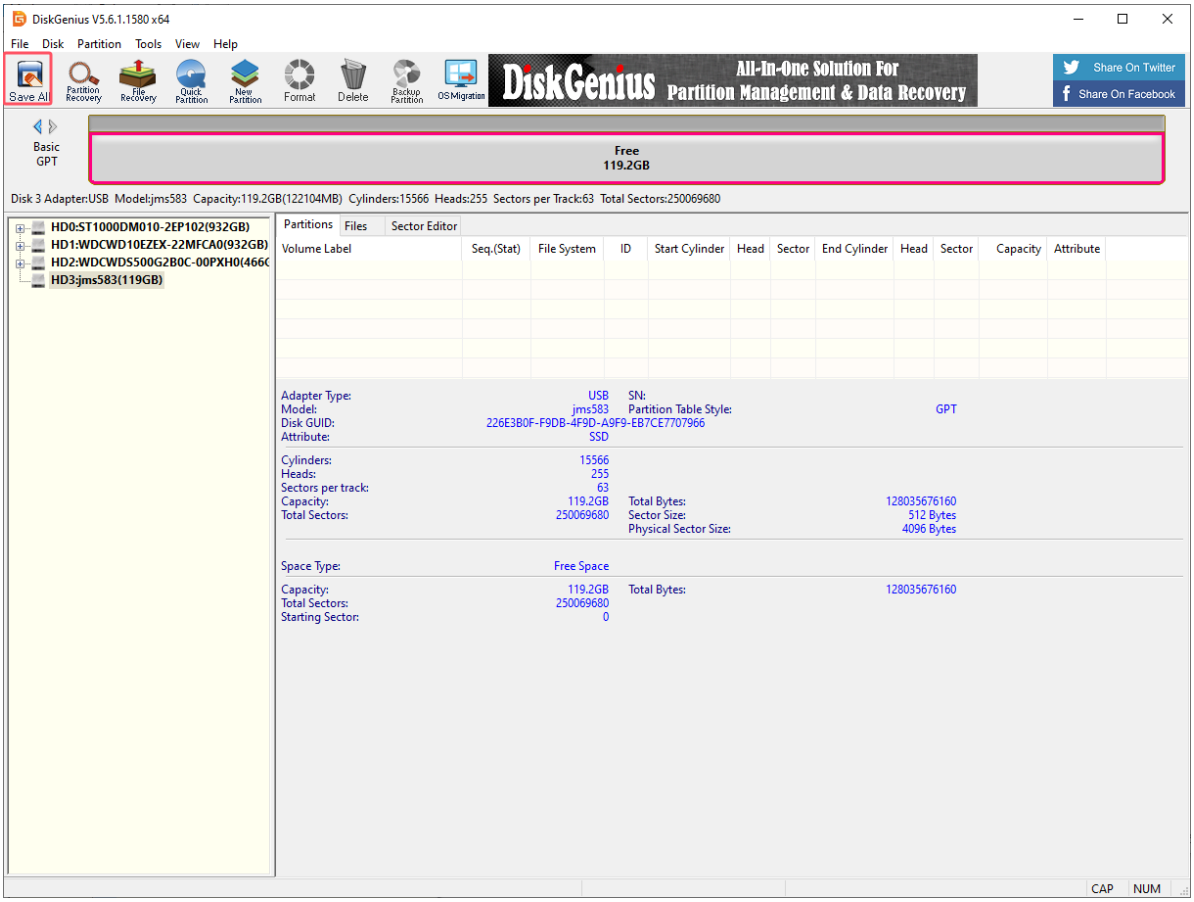
1.1. Use DiskGenius

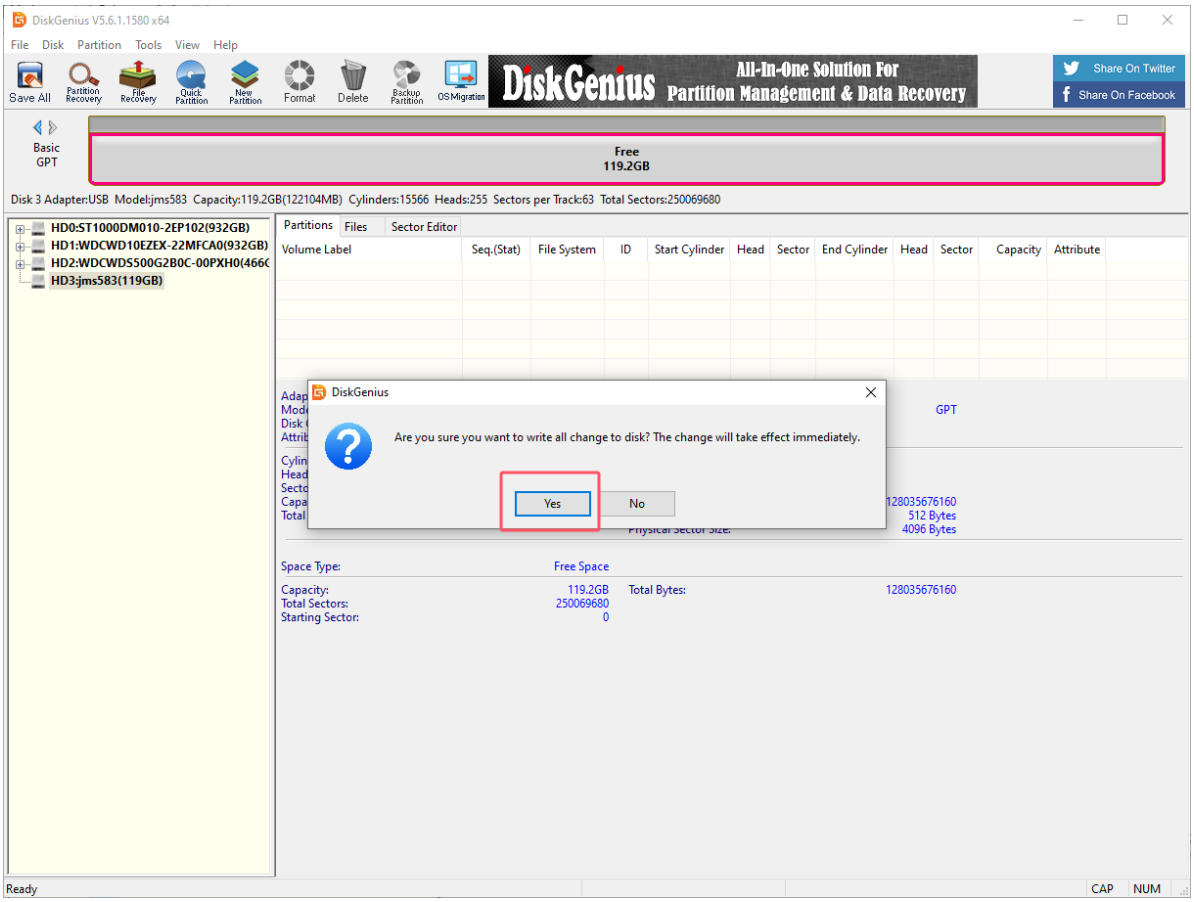
1.1.1, Delete partition

Deleting a partition will clear the disk data. Please confirm whether the drive letter is the disk that needs to be formatted before confirming the operation: you can judge based on the disk size and the newly added drive letter of the connected disk


DiskGenius V5.6.1.1580 x64
File Disk Partition Tools View Help
Save All
Partition Recovery
File Recovery
Quick Partition
New Partition
Format
Delete
Backup Partition
OS Migration
DiskGenius All-In-One Solution For Partition Management & Data Recovery
Share On Twitter
Share On Facebook
Basic GPT
Local Disk(T:)
EXT4
117.8GB
Disk 3 Adapter:USB Model:jms583 Capacity:119.2GB(122104MB) Cylinders:15566 Heads:255 Sectors per Track:63 Total Sectors:250069680
HD0:ST1000DM010-2EP102(932GB)
HD1:WDCWD10EZEX-22MFCA0(932GB)
HD2:WDCWDS500G2B0C-00PXH0(466
Partitions Files Sector Editor
Volume Label Seq.(Stat) File System ID Start Cylinder Head Sector End Cylinder Head Sector Capacity Attribute
Partition(E:) 0 MS ... 0 0 41 16 81 41 128.0MB
1 MS ... 16 81 42 16 106 2 768.0KB
2 MS ... 16 106 3 20 114 6 31.6MB
3 MS ... 20 114 7 36 195 7 128.0MB
4 MS ... 36 195 8 36 219 31 768.0KB
5 MS ... 36 219 32 40 227 35 31.6MB
Save Partition Table(F8)
Backup Partition Table(F2)
Restore Partition Table(F3)
Search Lost Partitions(Partition Recovery)(L)
Recover Lost Files(U)
Rebuild Master Boot Record(M)
Clear Reserved Sectors(E)
Quick Partition(F6)
Delete All Partitions(A)
Erase Sectors(E)
Backup Disk To Image File
Restore Image File To Disk
Convert To GUID Partition Table (P)
Convert To MBR Partition Table (B)
Convert To Basic Disk (D)
Appoint Disk Geometry(G)
Modify MBR Signature / Disk GUID
View S.M.A.R.T. Information(S)
View Storage Pool Information
Verify Or Repair Bad Sectors(Y)
Reset Bad Sector Records
Convert Boot Mode
Change Device State
TRIM Optimization
Close Virtual Disk File(C)
Recompose Virtual RAID
USB
jms583
226E3B0F-F9DB-4F9D-A9F9-EB7CE7707966
SSD
SN:
Partition Table Style: GPT
15566
255
63
119.2GB
250069680
Total Bytes: 128035676160
Sector Size: 512 Bytes
Physical Sector Size: 4096 Bytes
CAP NUM


DiskGenius V5.6.1.1580 x64
File Disk Partition Tools View Help
Save All
Partition Recovery
File Recovery
Quick Partition
New Partition
Format
Delete
Backup Partition
OS Migration
DiskGenius All-In-One Solution For Partition Management & Data Recovery
Share On Twitter
Share On Facebook
Basic GPT
Local Disk(T:)
EXT4
117.8GB
Disk 3 Adapter:USB Model:jms583 Capacity:119.2GB(122104MB) Cylinders:15566 Heads:255 Sectors per Track:63 Total Sectors:250069680
HD0:ST1000DM010-2EP102(932GB)
HD1:WDCWD10EZEX-22MFCA0(932GB)
HD2:WDCWDS500G2B0C-00PXH0(466
HD3:jms583(119GB)
Partition(E:)
Partition(H:)
Partition(I:)
Partition(J:)
Partition(K:)
Partition(L:)
Partition(M:)
Partition(N:)
ESP(8)
Partition(O:)
Partition(P:)
Partition(Q:)
Partition(R:)
Partition(S:)
Local Disk(T:)
Partitions Files Sector Editor
Volume Label Seq.(Stat) File System ID Start Cylinder Head Sector End Cylinder Head Sector Capacity Attribute
Partition(E:) 0 MS ... 0 0 41 16 81 41 128.0MB
Partition(H:) 1 MS ... 16 81 42 16 106 2 768.0KB
Partition(I:) 2 MS ... 16 106 3 20 114 6 31.6MB
Partition(J:) 3 MS ... 20 114 7 36 195 7 128.0MB
Partition(K:) 4 MS ... 36 195 8 36 219 31 768.0KB
Partition(L:) 5 MS ... 36 219 32 40 227 35 31.6MB
DiskGenius
Are you sure you want to delete all the partitions on current disk? All files in it will lose. Select "Yes" to delete.
Yes
No
Ready
CAP NUM

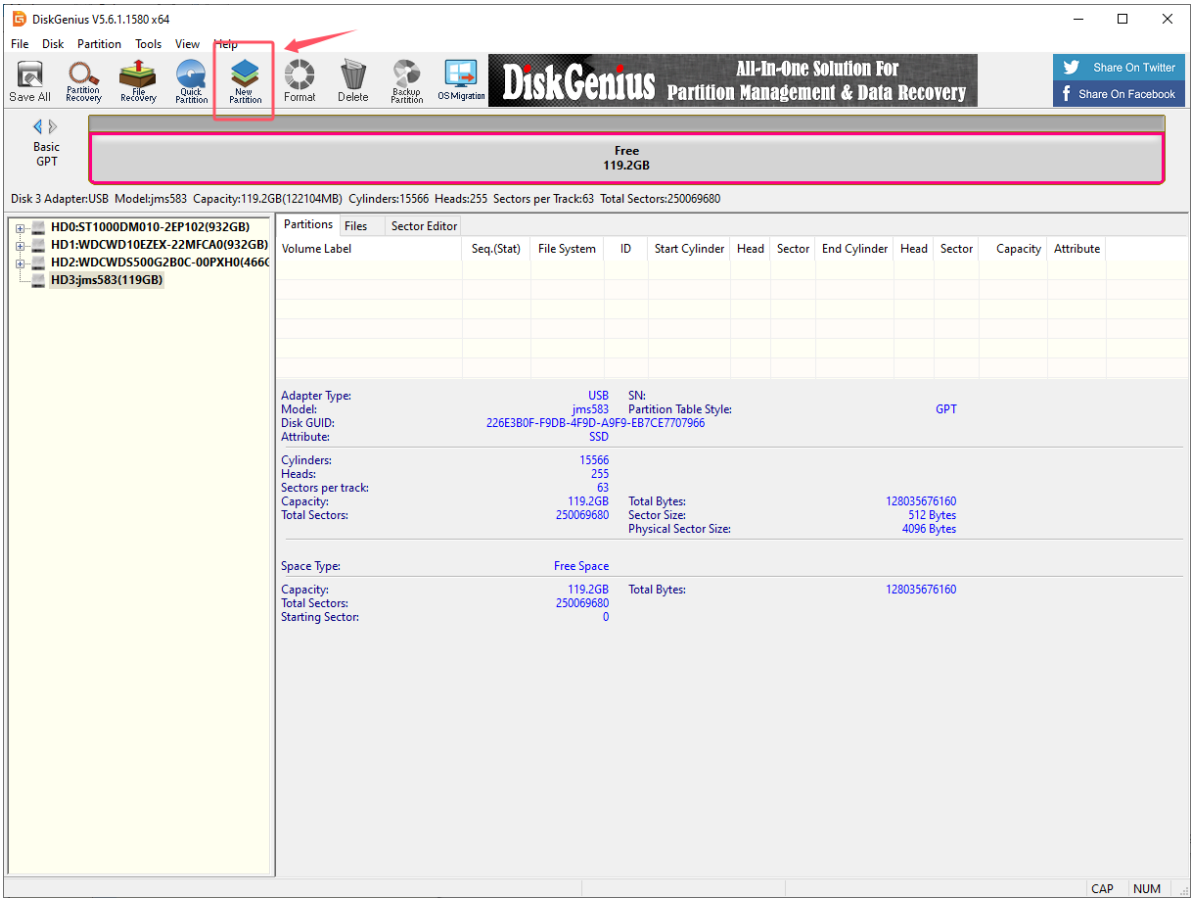


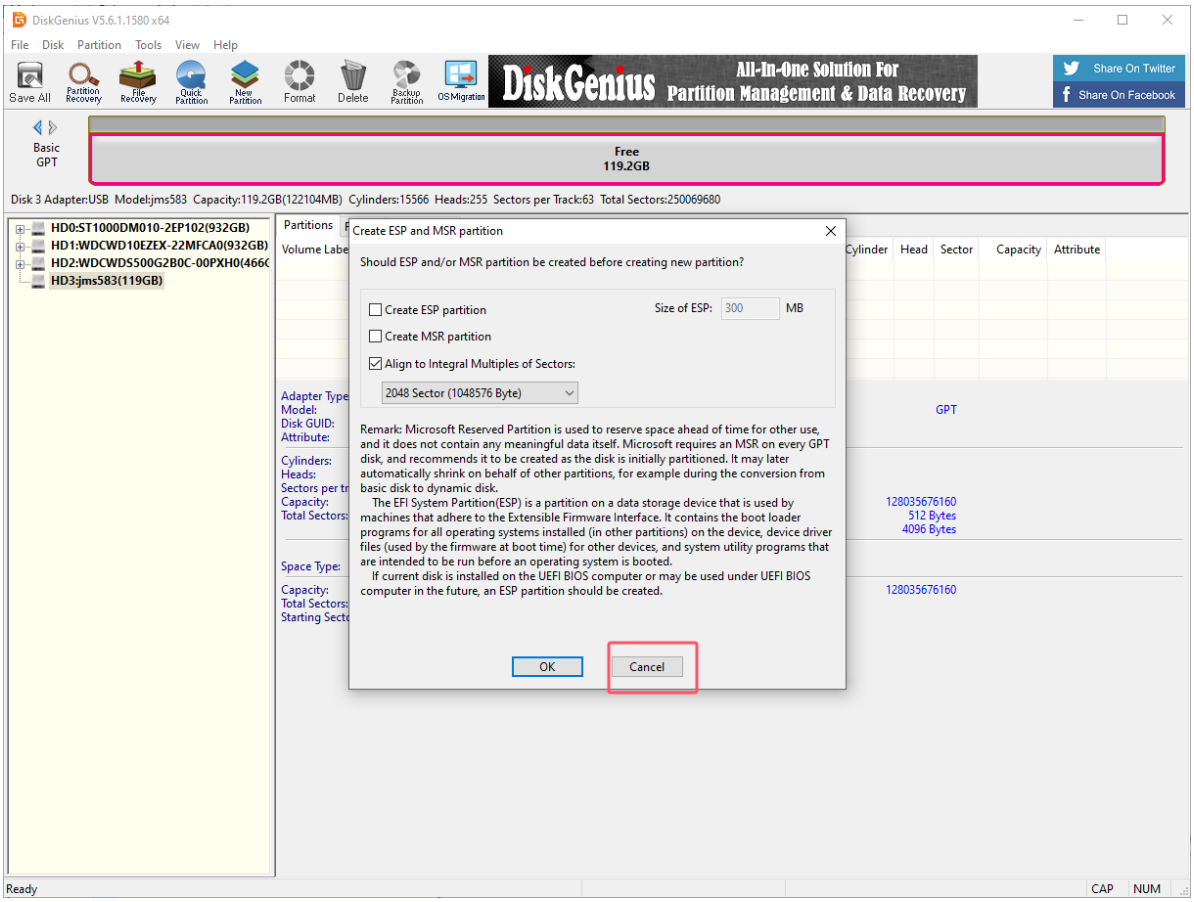


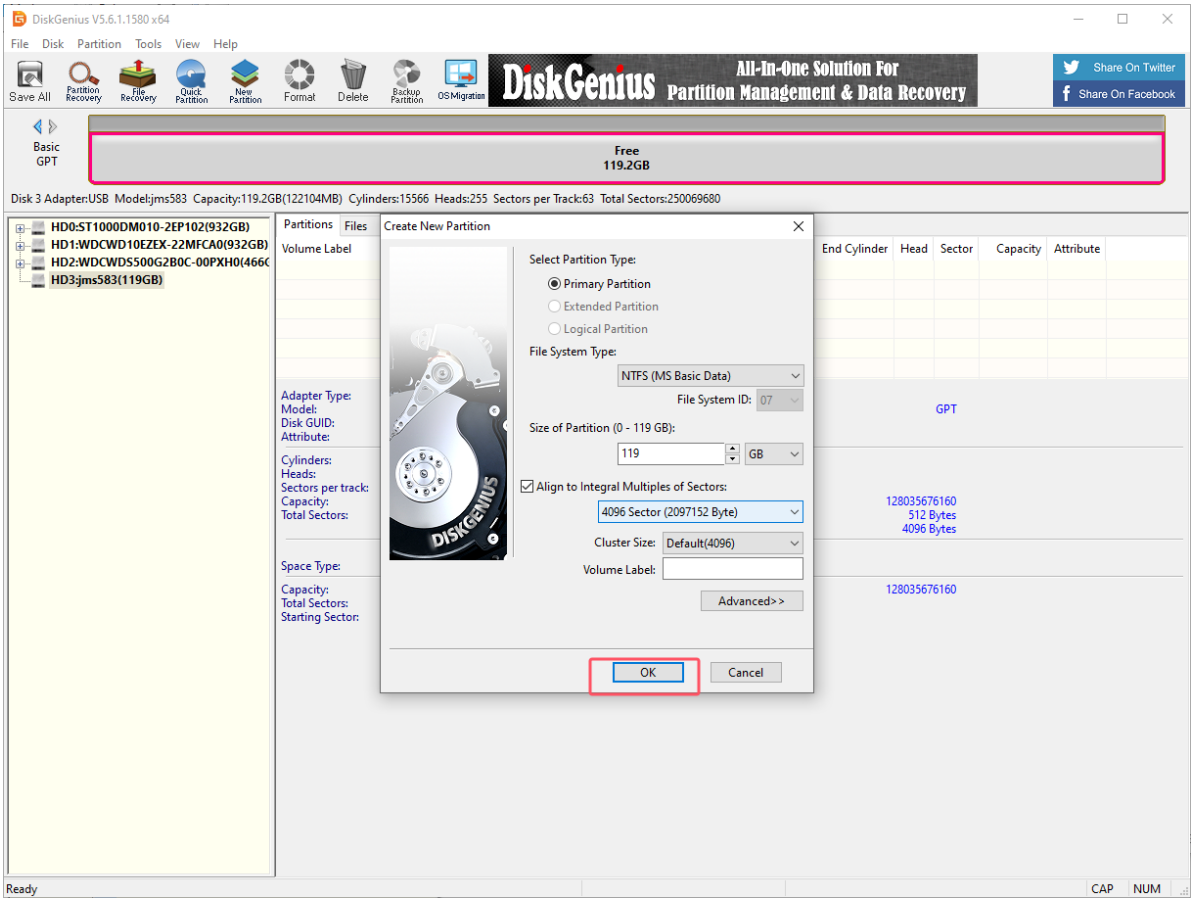
1.1.2. Create a new partition

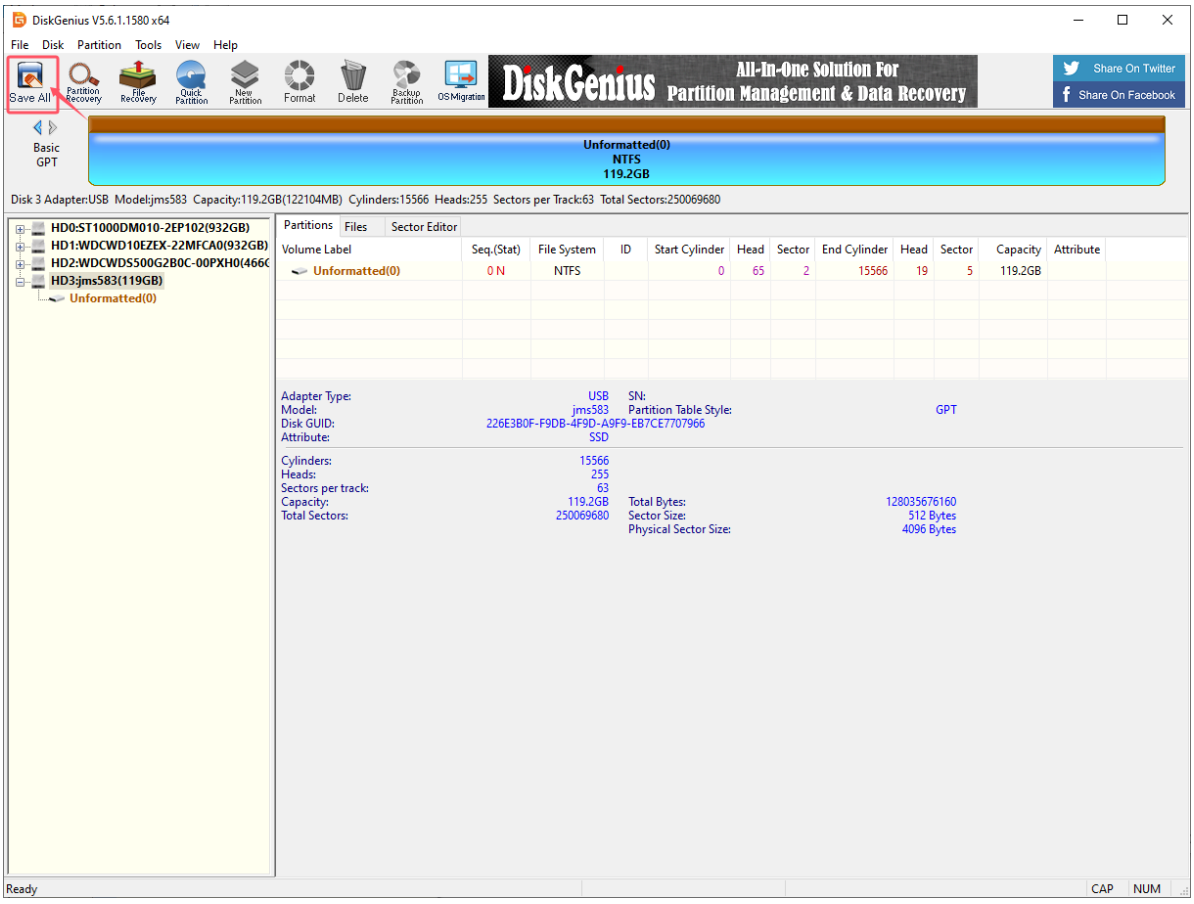
Partition the SSD into NTFS format.

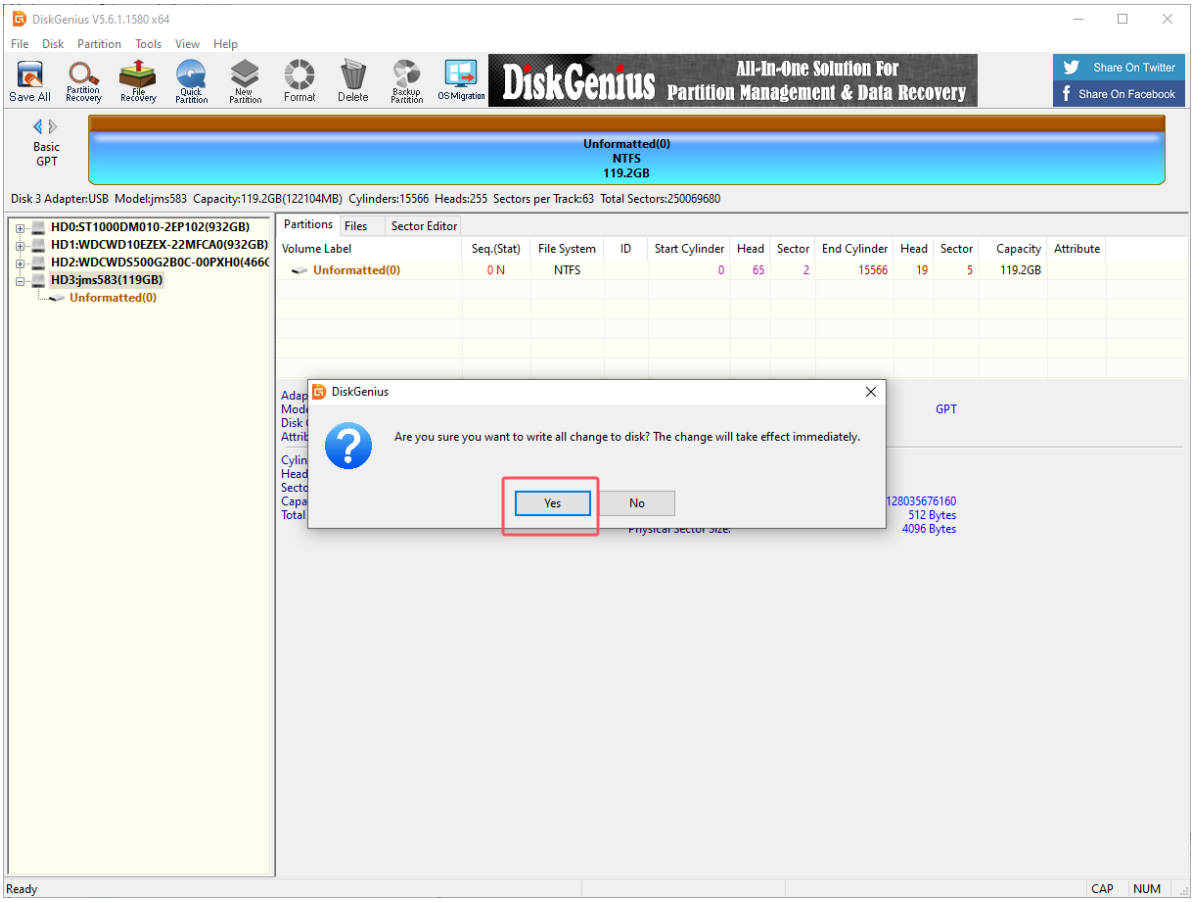
Select the drive letter corresponding to the SSD, and then click New Partition:

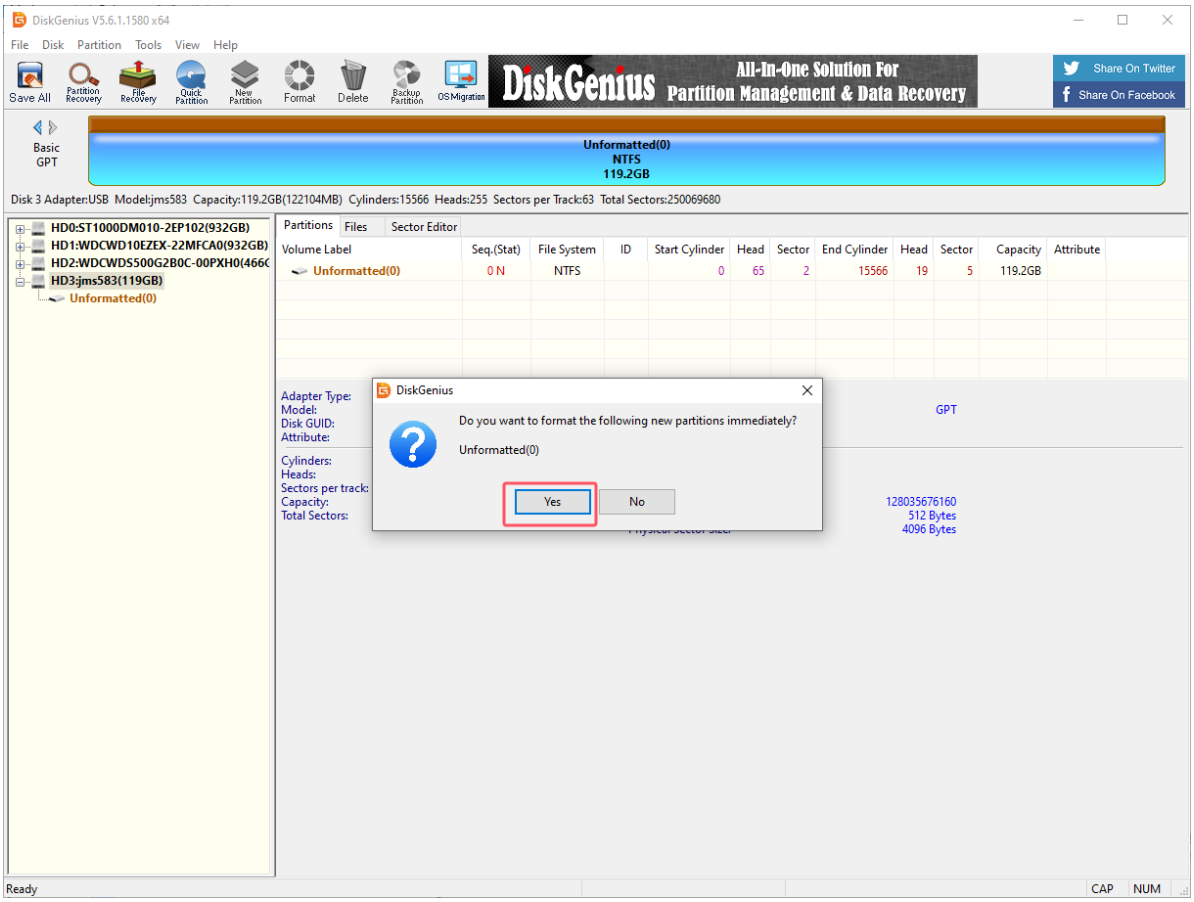

DiskGenius V5.6.1.1580 x64
File Disk Partition Tools View Help
Save All
Partition Recovery
File Recovery
Quick Partition
New Partition
Format
Delete
Backup Partition
OS Migration
DiskGenius All-In-One Solution For Partition Management & Data Recovery
Share On Twitter
Share On Facebook
Basic GPT
Free 119.2GB
Disk 3 Adapter:USB Model:jms583 Capacity:119.2GB(122104MB) Cylinders:15566 Heads:255 Sectors per Track:63 Total Sectors:250069680
HD0:ST1000DM010-2EP102(932GB)
HD1:WDCWD10EZEX-22MFCA0(932GB)
HD2:WDCWDS500G2B0C-00PXH0(466(
HD3:jms583(119GB)
Partitions Files Sector Editor
Volume Label Seq.(Stat) File System ID Start Cylinder Head Sector End Cylinder Head Sector Capacity Attribute
Adapter Type: USB
Model: jms583
Disk GUID: 226E3B0F-F9DB-4F9D-A9F9-EB7CE7707966
Attribute: SSD
SN:
Partition Table Style: GPT
Cylinders: 15566
Heads: 255
Sectors per track: 63
Capacity: 119.2GB
Total Sectors: 250069680
Total Bytes: 128035676160
Sector Size: 512 Bytes
Physical Sector Size: 4096 Bytes
Space Type: Free Space
Capacity: 119.2GB
Total Sectors: 250069680
Starting Sector: 0
Total Bytes: 128035676160
CAP NUM

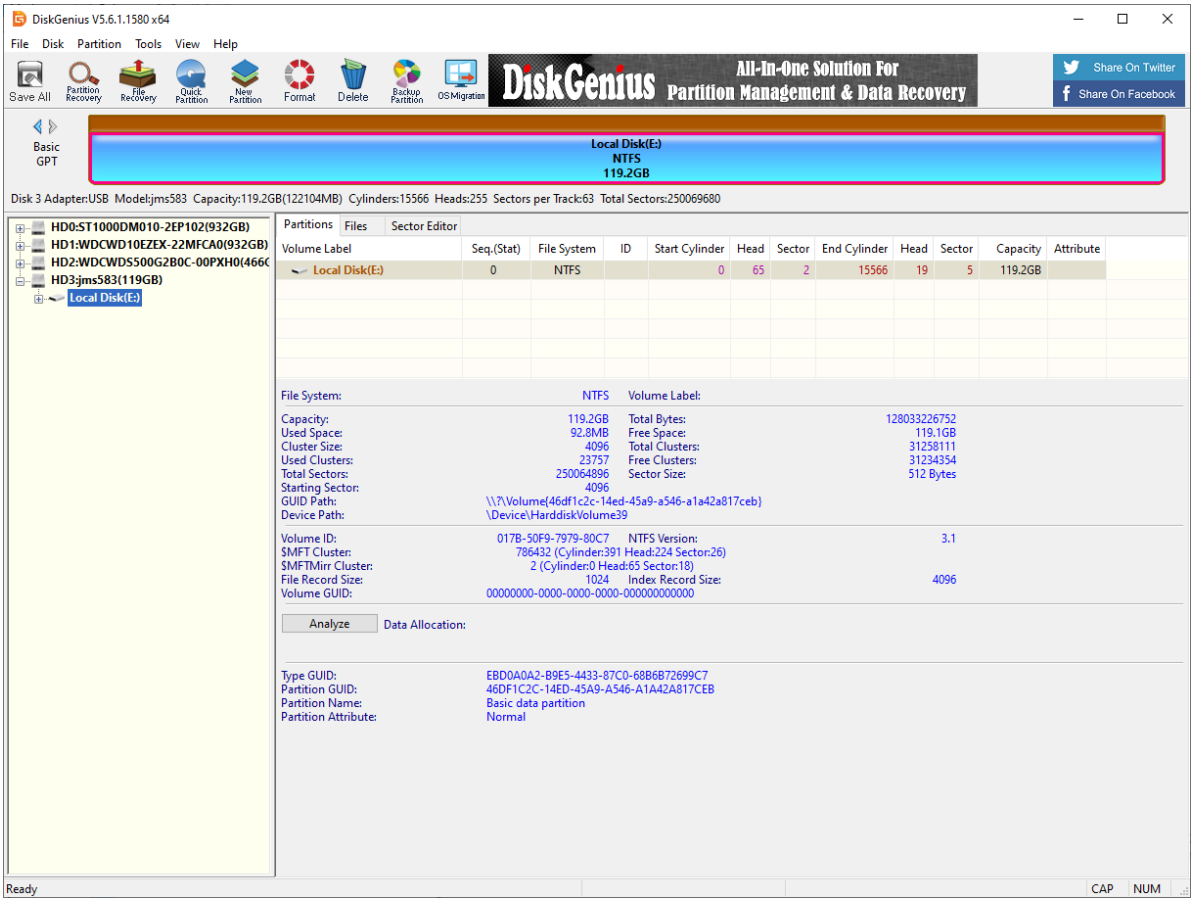

DiskGenius V5.6.1.1580 x64
File Disk Partition Tools View Help
Save All
Partition Recovery
File Recovery
Quick Partition
New Partition
Format
Delete
Backup Partition
OS Migration
DiskGenius All-In-One Solution For Partition Management & Data Recovery
Share On Twitter
Share On Facebook
Basic GPT
Free 119.2GB
Disk 3 Adapter:USB Model:jms583 Capacity:119.2GB(122104MB) Cylinders:15566 Heads:255 Sectors per Track:63 Total Sectors:250069680
HD0:ST1000DM010-2EP102(932GB)
HD1:WDCWD10EZEX-22MFCA0(932GB)
HD2:WDCWDS500G2B0C-00PXH0(466(
HD3:jms583(119GB)
Create ESP and MSR partition
Should ESP and/or MSR partition be created before creating new partition?
Create ESP partition
Size of ESP: 300 MB
Create MSR partition
Align to Integral Multiples of Sectors:
2048 Sector (1048576 Byte)
Remark: Microsoft Reserved Partition is used to reserve space ahead of time for other use, and it does not contain any meaningful data itself. Microsoft requires an MSR on every GPT disk, and recommends it to be created as the disk is initially partitioned. It may later automatically shrink on behalf of other partitions, for example during the conversion from basic disk to dynamic disk.
The EFI System Partition(ESP) is a partition on a data storage device that is used by machines that adhere to the Extensible Firmware Interface. It contains the boot loader programs for all operating systems installed (in other partitions) on the device, device driver files (used by the firmware at boot time) for other devices, and system utility programs that are intended to be run before an operating system is booted.
If current disk is installed on the UEFI BIOS computer or may be used under UEFI BIOS computer in the future, an ESP partition should be created.
OK
Cancel
Ready
CAP NUM


DiskGenius V5.6.1.1580 x64
File Disk Partition Tools View Help
Save All
Partition Recovery
File Recovery
Quick Partition
New Partition
Format
Delete
Backup Partition
OS Migration
DiskGenius All-In-One Solution For Partition Management & Data Recovery
Share On Twitter
Share On Facebook
Basic GPT
Free
119.2GB
Disk 3 Adapter:USB Model:jms583 Capacity:119.2GB(122104MB) Cylinders:15566 Heads:255 Sectors per Track:63 Total Sectors:250069680
HD0:ST1000DM010-2EP102(932GB)
HD1:WDCWD10EZEX-22MFCA0(932GB)
HD2:WDCWDS500G2B0C-00PXH0(466
HD3:jms583(119GB)
Partitions Files
Volume Label
End Cylinder Head Sector Capacity Attribute
Create New Partition
Select Partition Type:
Primary Partition
Extended Partition
Logical Partition
File System Type:
NTFS (MS Basic Data)
File System ID: 07
Size of Partition (0 - 119 GB):
119 GB
Align to Integral Multiples of Sectors:
4096 Sector (2097152 Byte)
Cluster Size: Default(4096)
Volume Label:
Advanced>>
OK
Cancel
Adapter Type:
Model:
Disk GUID:
Attribute:
GPT
Cylinders:
Heads:
Sectors per track:
Capacity:
Total Sectors:
128035676160
512 Bytes
4096 Bytes
Space Type:
Capacity:
Total Sectors:
Starting Sector:
128035676160
Ready
CAP NUM


DiskGenius V5.6.1.1580 x64
File Disk Partition Tools View Help
Save All
Partition Recovery
File Recovery
Quick Partition
New Partition
Format
Delete
Backup Partition
OS Migration
DiskGenius All-In-One Solution For Partition Management & Data Recovery
Share On Twitter
Share On Facebook
Basic GPT
Unformatted(0)
NTFS
119.2GB
Disk 3 Adapter:USB Model:jms583 Capacity:119.2GB(122104MB) Cylinders:15566 Heads:255 Sectors per Track:63 Total Sectors:250069680
HD0:ST1000DM010-2EP102(932GB)
HD1:WDCWD10EZEX-22MFCA0(932GB)
HD2:WDCWDS500G2B0C-00PXH0(466
HD3:jms583(119GB)
Unformatted(0)
Partitions Files Sector Editor
Volume Label Seq.(Stat) File System ID Start Cylinder Head Sector End Cylinder Head Sector Capacity Attribute
Unformatted(0) 0 N NTFS 0 65 2 15566 19 5 119.2GB
Adapter Type: USB
Model: jms583
Disk GUID: 226E3B0F-F9DB-4F9D-A9F9-EB7CE7707966
Attribute: SSD
SN:
Partition Table Style: GPT
Cylinders: 15566
Heads: 255
Sectors per track: 63
Capacity: 119.2GB
Total Sectors: 250069680
Total Bytes: 128035676160
Sector Size: 512 Bytes
Physical Sector Size: 4096 Bytes
Ready
CAP NUM


DiskGenius V5.6.1.1580 x64
File Disk Partition Tools View Help
Save All
Partition Recovery
File Recovery
Quick Partition
New Partition
Format
Delete
Backup Partition
OS Migration
DiskGenius
All-In-One Solution For
Partition Management & Data Recovery
Share On Twitter
Share On Facebook
Basic
GPT
Unformatted(0)
NTFS
119.2GB
Disk 3 Adapter:USB Model:jms583 Capacity:119.2GB(122104MB) Cylinders:15566 Heads:255 Sectors per Track:63 Total Sectors:250069680
HD0:ST1000DM010-2EP102(932GB)
HD1:WDCWD10EZEX-22MFCA0(932GB)
HD2:WDCWDS500G2B0C-00PXH0(466
HD3:jms583(119GB)
Unformatted(0)
Partitions Files Sector Editor
Volume Label Seq.(Stat) File System ID Start Cylinder Head Sector End Cylinder Head Sector Capacity Attribute
Unformatted(0) 0 N NTFS 0 65 2 15566 19 5 119.2GB
DiskGenius
Are you sure you want to write all change to disk? The change will take effect immediately.
Yes
No
GPT
128035676160
512 Bytes
4096 Bytes
Ready
CAP
NUM


DiskGenius V5.6.1.1580 x64
File Disk Partition Tools View Help
Save All
Partition Recovery
File Recovery
Quick Partition
New Partition
Format
Delete
Backup Partition
OS Migration
DiskGenius
All-In-One Solution For
Partition Management & Data Recovery
Share On Twitter
Share On Facebook
Basic
GPT
Unformatted(0)
NTFS
119.2GB
Disk 3 Adapter:USB Model:jms583 Capacity:119.2GB(122104MB) Cylinders:15566 Heads:255 Sectors per Track:63 Total Sectors:250069680
HD0:ST1000DM010-2EP102(932GB)
HD1:WDCWD10EZEX-22MFCA0(932GB)
HD2:WDCWDS500G2B0C-00PXH0(466
HD3:jms583(119GB)
Unformatted(0)
Partitions Files Sector Editor
Volume Label Seq.(Stat) File System ID Start Cylinder Head Sector End Cylinder Head Sector Capacity Attribute
Unformatted(0) 0 N NTFS 0 65 2 15566 19 5 119.2GB
Adapter Type:
Model:
Disk GUID:
Attribute:
Cylinders:
Heads:
Sectors per track:
Capacity:
Total Sectors:
DiskGenius
Do you want to format the following new partitions immediately?
Unformatted(0)
Yes
No
GPT
128035676160
512 Bytes
4096 Bytes
Ready
CAP
NUM

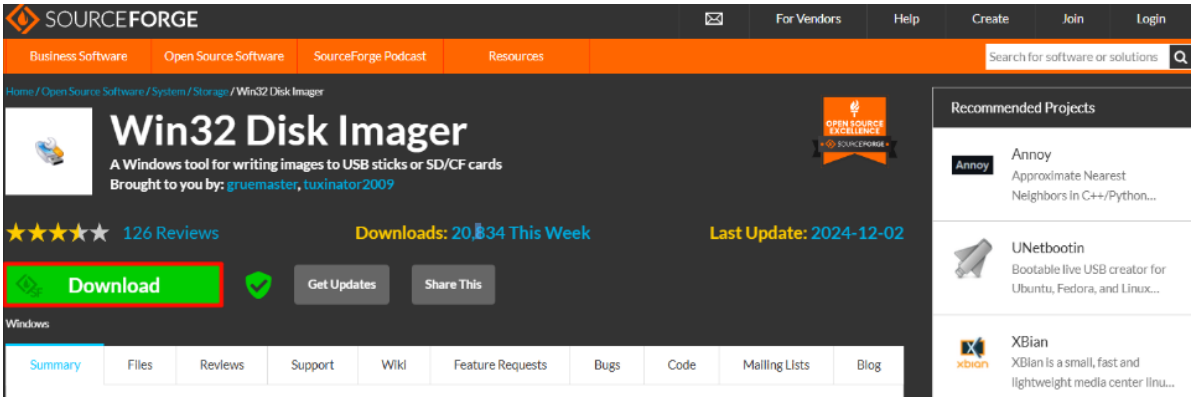


2. Restore the factory image

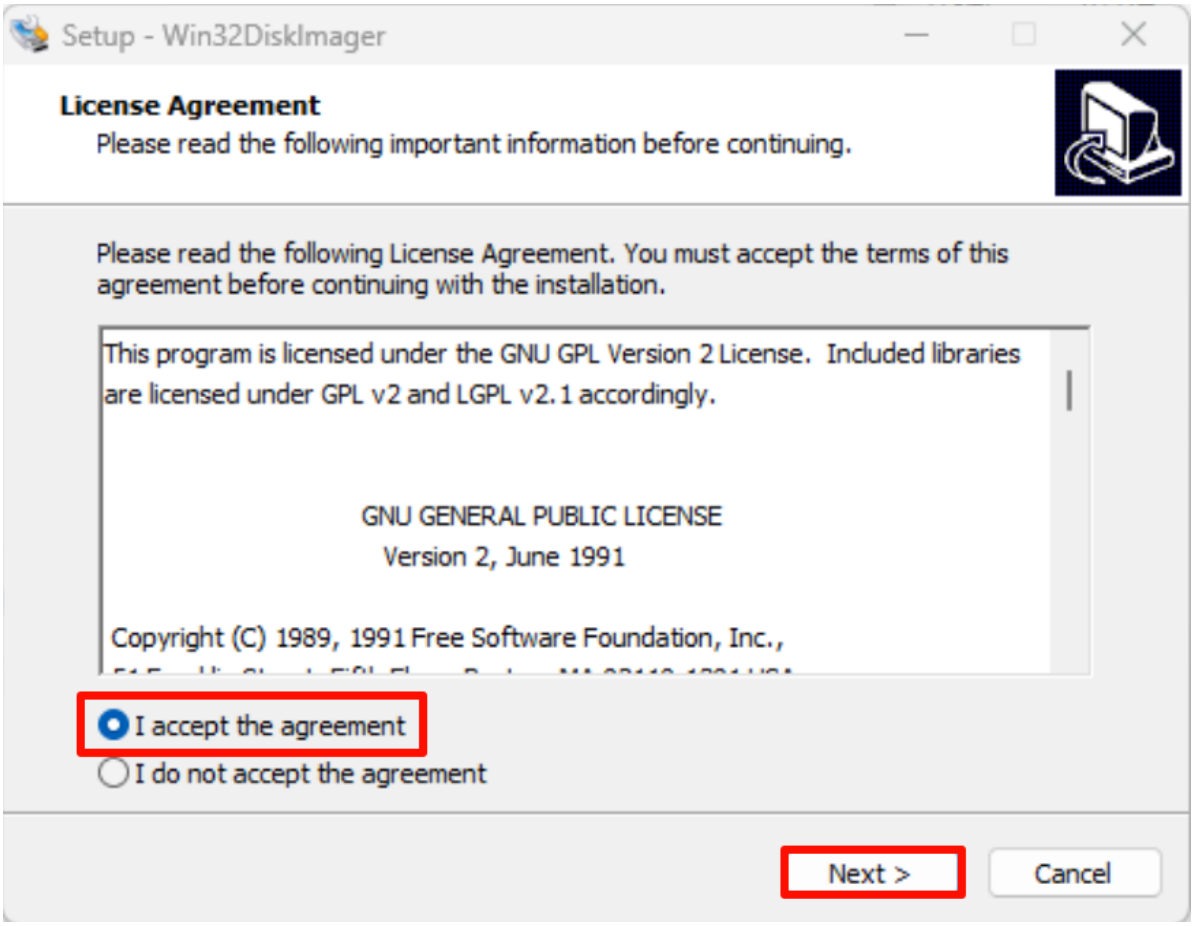
You need to download and decompress the factory image system in the data to the local computer in advance.

2.1. Install Win32DiskImager

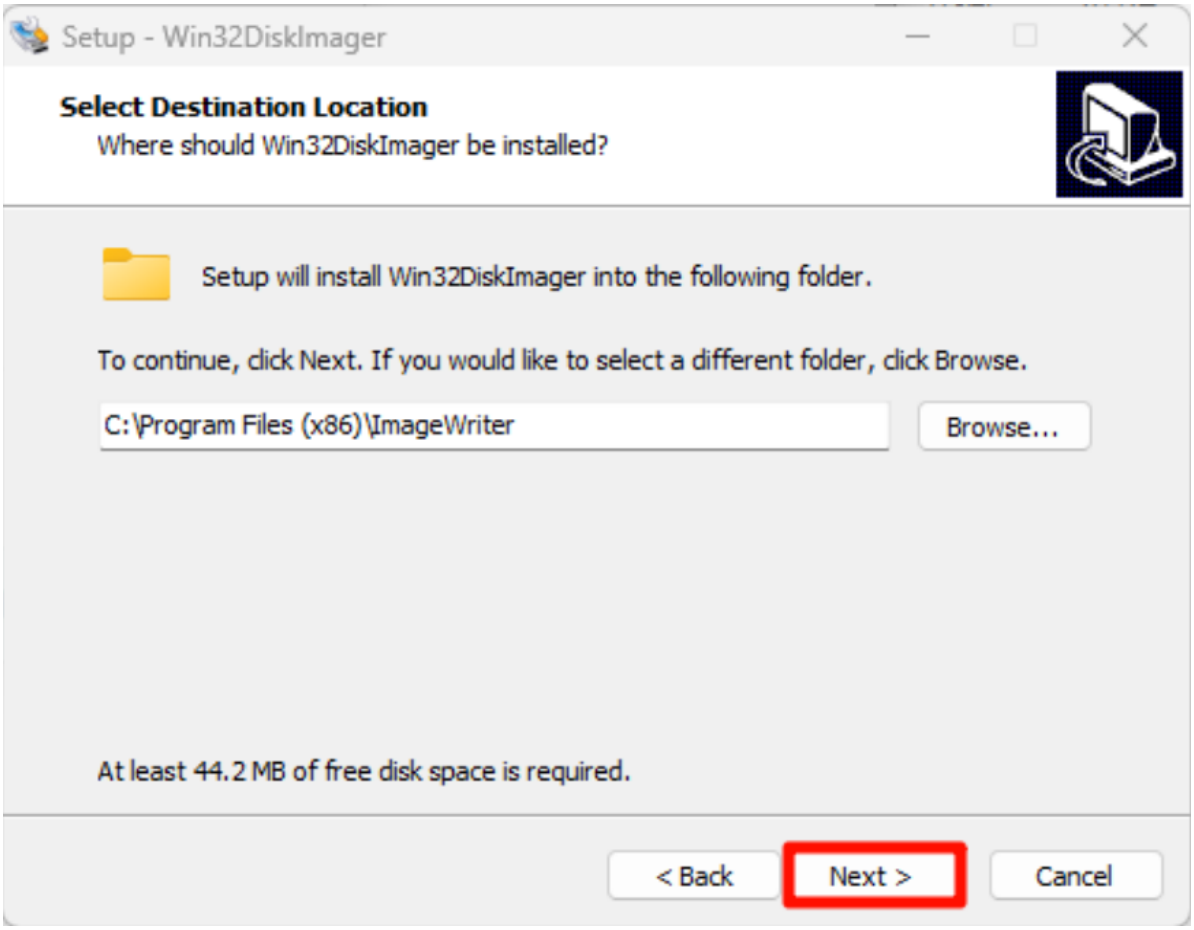
Download URL: https://sourceforge.net/projects/win32diskimager/



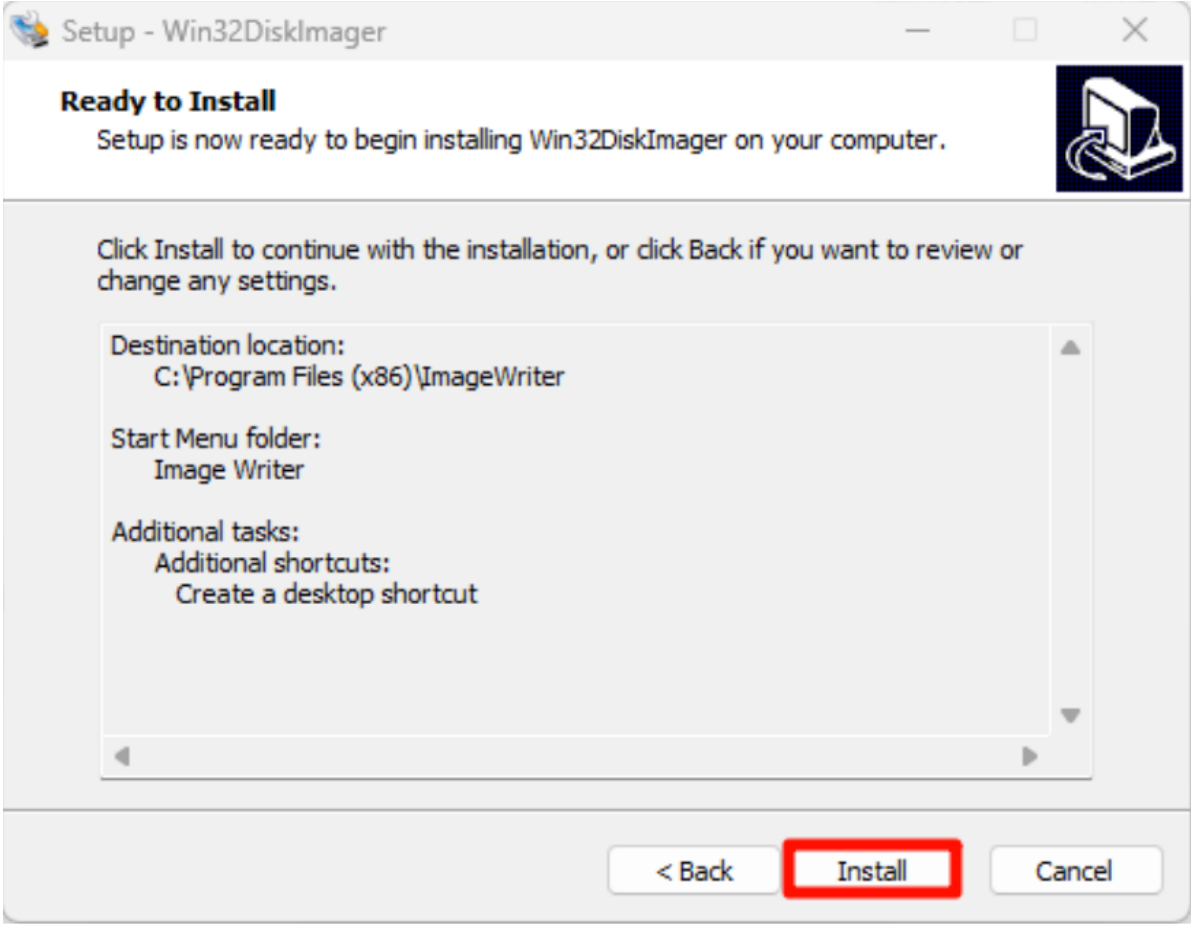
Open the `win32diskimager-1.0.0-install.exe` installation package as an administrator and accept the agreement:

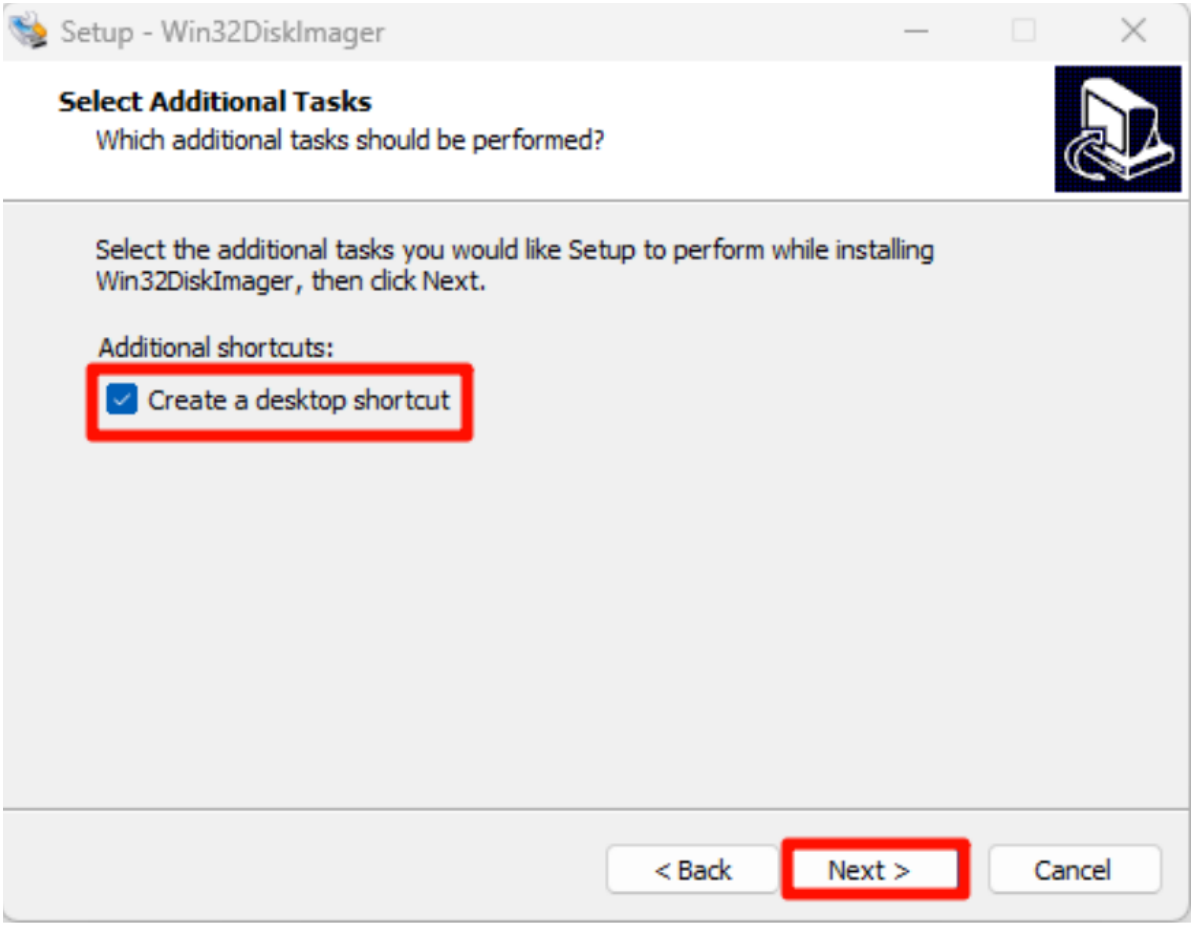


Installation location: The default location is recommended

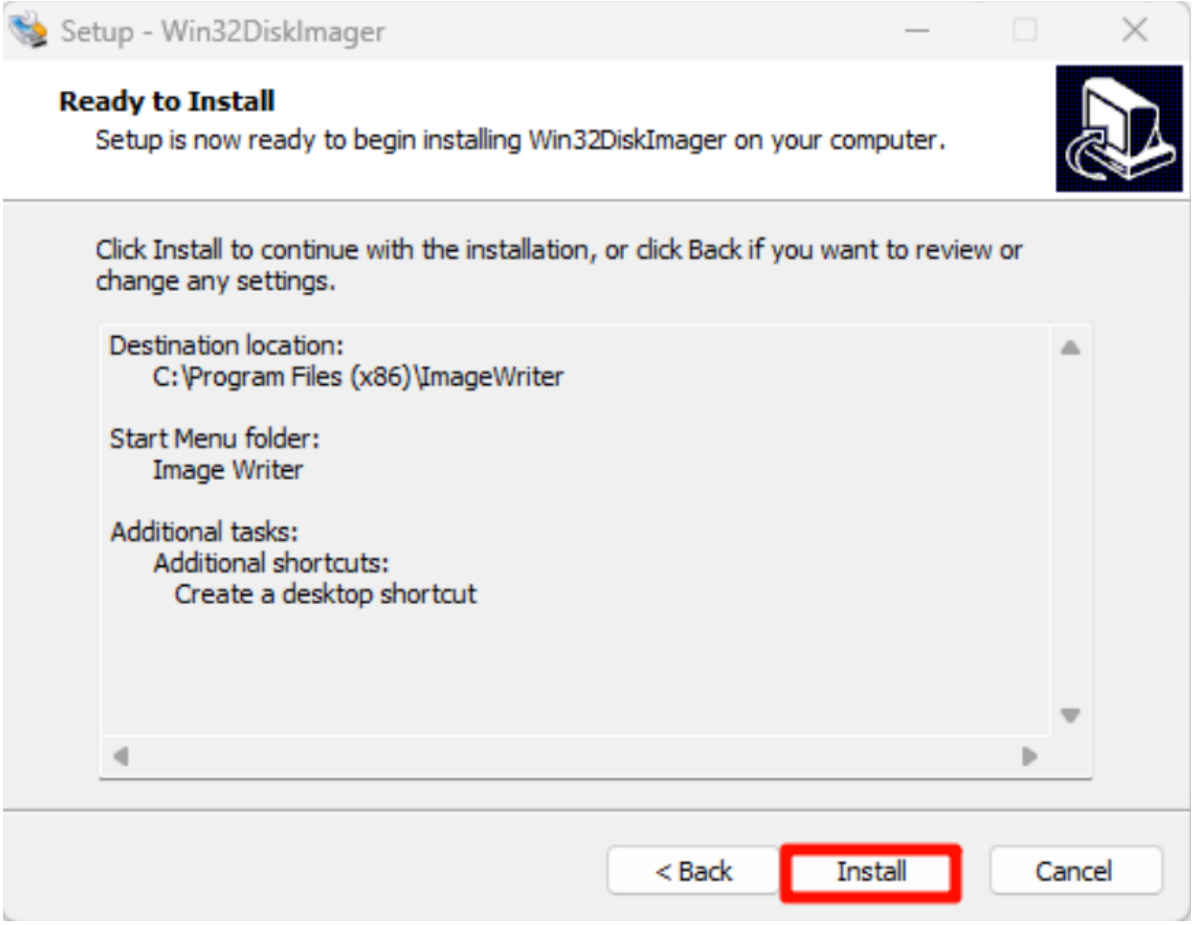


Installation options:

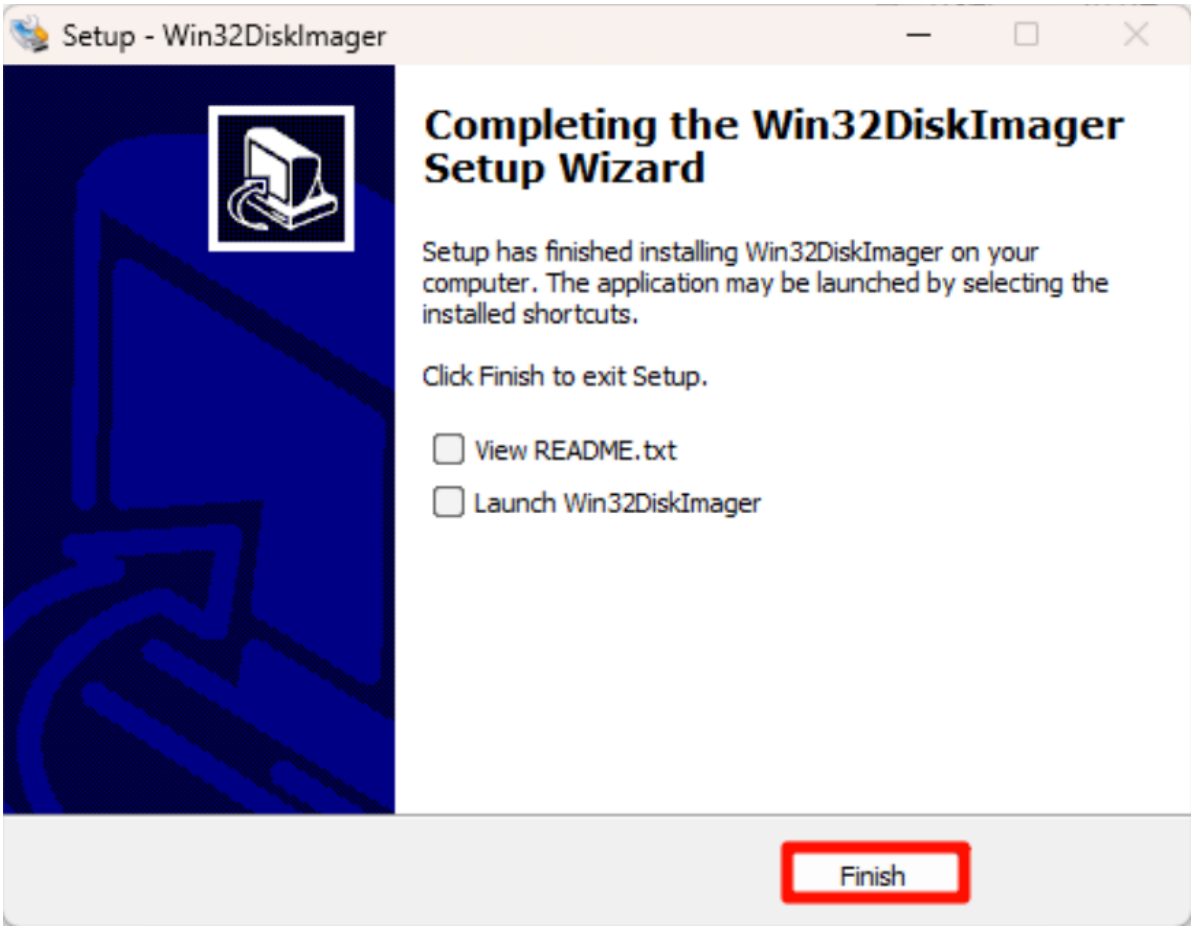




Start installation:



Complete installation:

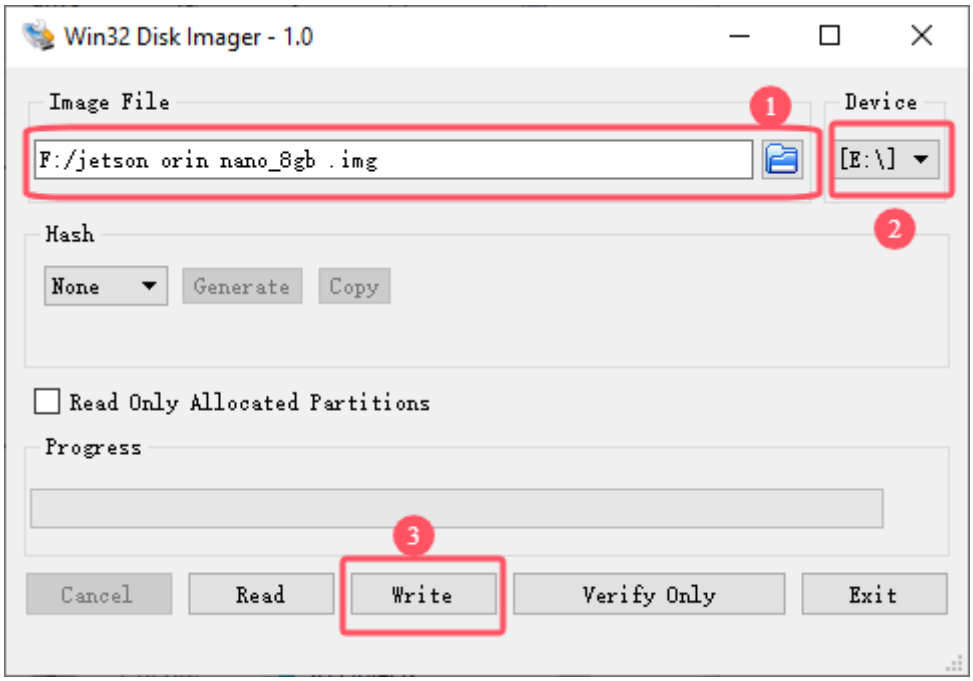


2.2. Use Win32DiskImager

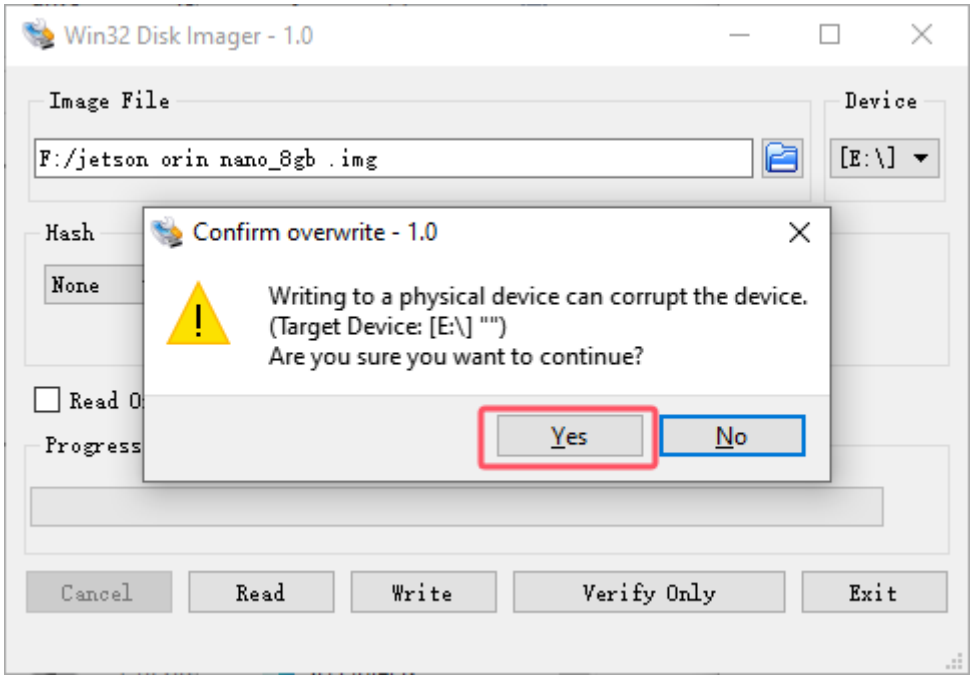
①: Select the factory image file (*.img) in the data

②: Select the drive letter corresponding to the solid-state drive

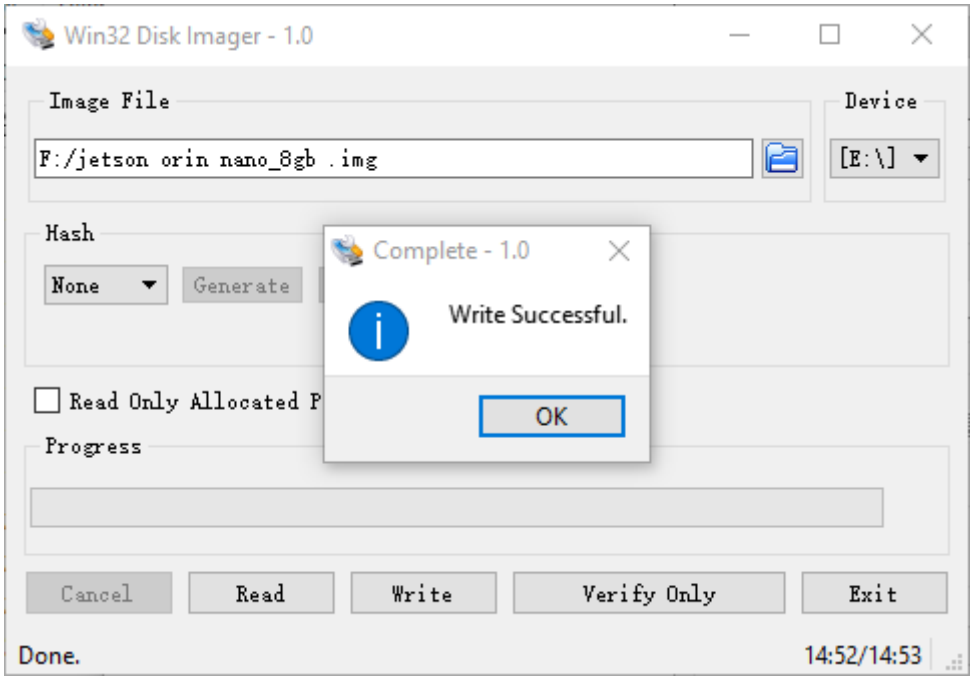
③: Write the factory image to the solid-state drive



Confirm writing to the system:



Wait for the system to be written successfully:



After the system is written, you can close the program and install the SSD to the Jetson AGX Thor motherboard!

3. Description

The Jetson motherboard can start the system normally and it depends on the system Jetpack version. Generally, only the same version can start the system!